

Stopping-Target Foils vs Electron-Beam Flash

Extra foils swing tracker flash per POT $3.1\times$ — at no $S/\sqrt{B}$ cost

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Mu2e — autoresearch / closed-loop Bayesian optimization

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Result in one slide

Extra stopping-target foils are a **real lever on the electron-beam flash**: tracker flash per POT varies **3.1×** across the explored geometry, at **no $S/\sqrt{B}$ cost**.

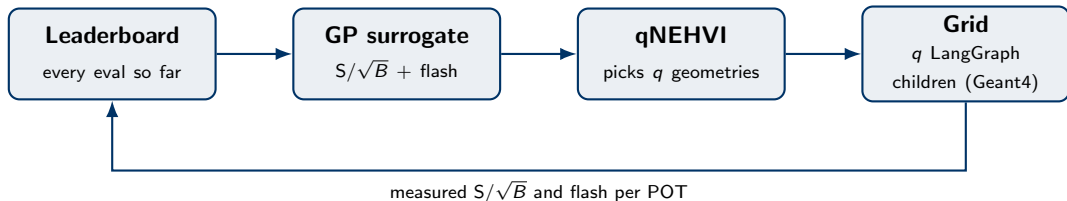
- **339 evals** from closed-loop Bayesian optimization map a clean Pareto front.
- **Champion** ff11R00_07 beats the deployed target on *both* axes: $S/\sqrt{B}$ 3.31 vs 3.11, flash −3%.
- The $S/\sqrt{B}$ ceiling (**3.90**) is **confirmed reproducible**; a dense basin scan found BASIN01_11 — same 3.90 at **−5.8% flash** vs the transplant (400-seed confirmed, $\sim 3.2\sigma$).

What we optimize

6-D extra-foil geometry: up to 6 upstream and 6 downstream extra foils on the pinned 37-foil base — outer radius, half-thickness, and hole fraction, independently up and down.

objective	direction	source
$S/\sqrt{B}$	maximize	Run1A CE significance
e^- early-flash edep per POT in tracker	minimize	elebeam_flash, DS-on

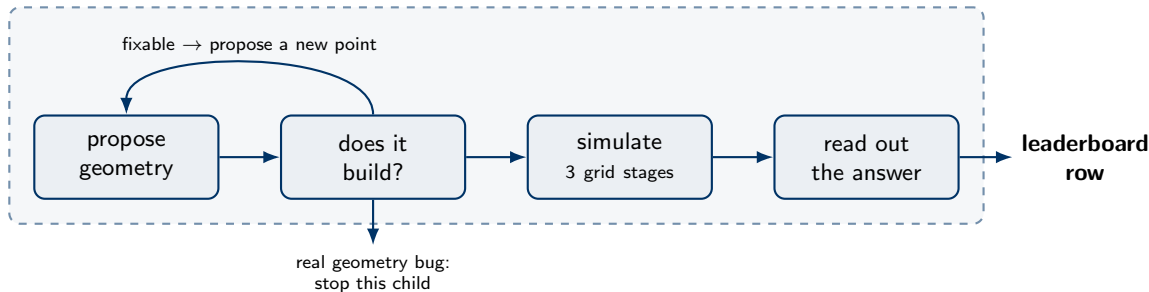
How the loop works



q geometries are in flight at once; each returns one measured row, and the surrogate refits as rows land. Nothing reaches the leaderboard that was not actually simulated.

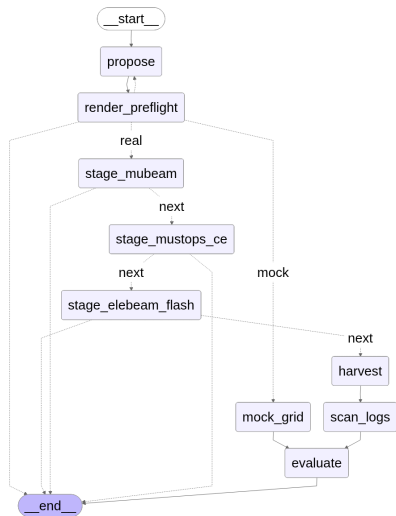
Inside one evaluation

one LangGraph run = one evaluation



- **LangGraph** is a small state-machine library: you declare *nodes* (steps) and *edges* (transitions); it runs them and saves the state after each.
- **We use it for exactly one job**: driving a single evaluation from proposal to leaderboard row — the optimizer loop around it is plain Python.
- Saved state means a child that dies mid-chain **resumes** instead of re-submitting grid jobs.
- **Read out the answer** = reduce the finished jobs' outputs to two numbers: CE tuple $\rightarrow S/\sqrt{B}$, tracker straw-gas energy $\rightarrow$ flash per POT.
- Any failed stage **halts the chain**: no row is ever written from partial results.

The actual LangGraph topology



Generated from the running code — `get_graph().draw_mermaid()`,
foilsflash mode — so it cannot drift from what actually executes.

- `propose / render_preflight` — pick the point, write the geometry, cheap local Geant4 init check.
- `stage_*` — the three grid stages, in chain order.
- `harvest / scan_logs` — reduce the outputs to the two metrics, and refuse the row if the log scan finds physics-breaking patterns.
- `evaluate` — append the leaderboard row.
- **Dotted** edges are conditional: the preflight retry, the per-stage fail-fast to `__end__`, and the mock path used by tests.
- **Control flow, not a timeline:** `stage_elebeam_flash` is last in the chain, but its jobs were submitted inside the `stage_mubeam` node — they ran during `stage_mustops_ce`, so the node only polls. The overlap is a side effect, not an edge.

Launch a campaign (*q* children, rolling):

```
AUTORESEARCH_MODE=foilsflash \  
python -m graph.closed_loop --mode foilsflash \  
--name-prefix foilsflash20 --picker hybrid \  
--q 5 --max-evals 10 --rolling
```

One evaluation (exactly the graph at left):

```
python -m graph.run --mode foilsflash --no-mock \  
--config-name ff20R00.00 --x-point <6 values>
```

How the machinery behaves

question	answer
Where is LangGraph?	One <i>evaluation</i> = one LangGraph run. The optimizer loop around it is plain Python.
When does a stage pass?	Quorum, not all jobs: 12/15 for mubeam and mustops_ce (0.8), 90/100 for elebeam_flash (0.9).
Does harvest start there?	No — it also waits for the <i>outputs</i> to land on /pnfs. Jobs leave the queue minutes before stage-out finishes.
Does a failed stage retry?	No. Preflight retries 3× (2 min, local); a stage has already spent ~90 min of grid — it stops and a human looks.
What ends a run early?	Geometry that will not build, 3 failed preflight retries, or any failed stage — each ends with no row rather than a bad one.
What is the mock path?	Synthetic metrics for testing the wiring with no grid. Campaigns hardcode <code>--no-mock</code> .

A row is never written from partial results — that is the one invariant behind all of the above.

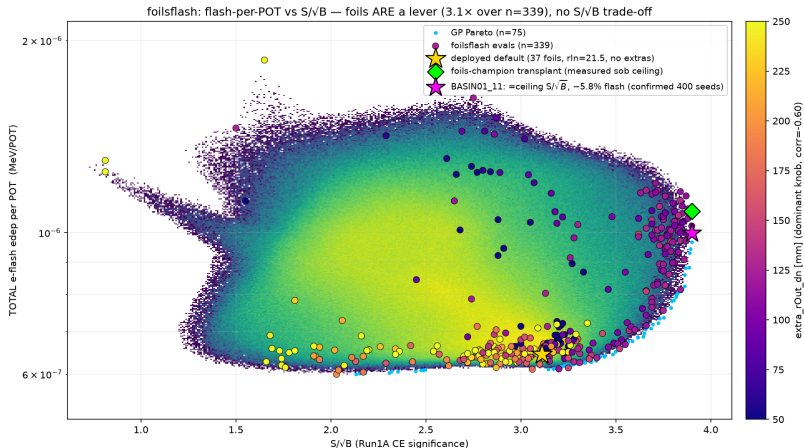
Picking the next q geometries

Same GP, different question asked of it — `--picker`:

picker	asks for	use when
hybrid	both, $\sim 60/40$	default — qNEHVI half mines the useful region, qN-ParEGO half patrols the tails
qnehvi	most new front area	mid-campaign, front still growing
qnparego	spread along the front	late campaign — keeps delivering when qNEHVI's hit-rate collapses
pareto_sob	highest predicted $S/\sqrt{B}$	exploiting the ceiling; flash still measured, so the corner gets <i>priced</i>
qlnei	$S/\sqrt{B}$ only	single-objective; drops flash as an objective

Every picker conditions on the evaluations already in flight (`X_pending`), so a batch never collides with itself or with running work. The 60/40 split is a knob: live attribution at deep saturation showed qNEHVI's front hit-rate falling $4/6 \rightarrow 0/6$ while qNParEGO kept scoring — so drop toward 0.3–0.4 at end of line.

The landscape



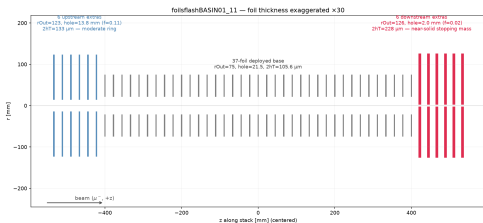
339 evaluations. Gold star = deployed default; green diamond = the transplant $S/\sqrt{B}$ ceiling; **magenta star** = BASIN01_11: same 3.90 ceiling at **-5.8% flash** (400-seed confirmed). Colour = downstream outer radius, the dominant knob. The cyan GP Pareto front traces the reachable low-flash / high- $S/\sqrt{B}$ edge.

The extra foils are a strong lever

- **3.1**× spread in flash per POT across the explored geometry (GP fit $R^2 \approx 0.9$).
- More **upstream** material → **more** flash; larger **downstream** radius with bigger holes → **less**.
- $S/\sqrt{B}$ and flash are nearly uncorrelated (-0.13) → cutting flash is essentially **free** of signal.

The foils act as a **scatterer of the on-axis electron beam** — the same picture behind Edmonds' central-hole result.

Anatomy of the $\max S/\sqrt{B}$ point



design	$S/\sqrt{B}$	flash [MeV/POT]
BASIN01_11 (confirmed)	3.90	1.00×10^{-6}
SOBX01 (transplant)	3.90	1.08×10^{-6}
ff14R01_00 (exploit)	3.84	8.14×10^{-7}
ff11R00_07 (champion)	3.31	6.26×10^{-7}
deployed default	3.11	6.45×10^{-7}

Two routes to high $S/\sqrt{B}$: a thin **off-beam** ring, or thin **in-beam** material that degrades the beam to stop more muons. A dense 20-point basin scan **confirms 3.90 reproduces** and finds BASIN01_11 at the ceiling with **−5.8% flash** vs the transplant (400-seed re-eval: 1.00 vs 1.06×10^{-6} , $\sim 3.2\sigma$). The flash floor itself is the **champion**, which beats deployed on both axes.

Top 10 configs by $S/\sqrt{B}$

Distinct geometries, replicates collapsed (flash averaged); 372 rows $\rightarrow$ 351 unique. Flash in 10^{-6} MeV/POT; $r_{\text{Out}}/hT/r_{\text{In}}$ in mm (up/dn); $r_{\text{In}} = f \cdot r_{\text{Out}}$ hole radius.

#	config	$S/\sqrt{B}$	flash	r_{Out}	hT	r_{In}	
1	BASIN01.00	3.90	1.075	112/110	0.063/0.145	20.0/0.0	transplant / ceiling
2	BASIN01.16	3.90	1.024	110/103	0.052/0.144	24.9/5.8	
3	BASIN01.11	3.90	0.999	123/126	0.066/0.114	13.8/2.0	
4	BASIN01.03	3.88	1.053	113/114	0.090/0.136	23.2/0.1	confirmed op. point
5	BASIN01.17	3.88	1.090	111/122	0.076/0.148	12.1/1.5	
6	BASIN01.04	3.87	1.142	91/112	0.044/0.177	9.5/3.7	
7	BASIN01.14	3.87	0.997	118/95	0.038/0.108	15.2/1.3	low-flash holed dn
8	BASIN01.09	3.86	0.971	104/117	0.024/0.123	26.6/4.1	
9	18R01.00	3.86	0.907	79/120	0.071/0.141	13.9/24.1	
10	BASIN01.02	3.86	1.042	131/98	0.045/0.162	31.7/2.2	
—	deployed (nominal)	3.11	0.645	—	—	—	37-foil base, no extras

Rows 1–3 all sit at the **3.90 ceiling** — BASIN01.11 owns it at the lowest flash. Row 9 (18R01.00) reaches **0.907** (downstream-holed) for only -0.04 in $S/\sqrt{B}$. All top configs run above the **0.645** nominal — the flash cost of the extra stopping mass that lifts $S/\sqrt{B}$ from 3.11 to ~ 3.9 . BASIN01-heavy: that scan densely mapped the high- $S/\sqrt{B}$ basin.

What the winning geometries share

The best- $S/\sqrt{B}$ configs converge on **compact foils with a nearly-solid downstream** and a $\sim$ deployed-size upstream hole — these are *stopping-target* foils, so more foil mass stops more muons.

parameter	top-10	median	
rIn_dn downstream hole	4.5 mm	48 mm	nearly solid
rIn_up upstream hole	19 mm	51 mm	$\sim$ deployed 21.5 mm
rOut up / dn [mm]	109 / 112	142 / 125	compact
hT up / dn [mm]	0.06 / 0.14	0.03 / 0.18	moderate

- **Every parameter correlates negatively with $S/\sqrt{B}$** — solid, compact foils add stopping mass: corr rIn_dn -0.61 , hT_dn -0.57 , rOut_dn -0.50 .
- **The flash lever is the downstream geometry:** the downstream hole rIn_dn (corr flash -0.64) and radius rOut_dn (-0.45) cut flash — and *both also cut $S/\sqrt{B}$* . That single tension is the Pareto front.
- **BASIN01_11 threads it:** it opened a small downstream hole (rIn_dn 2 mm) and pushed rOut_dn out (126 vs 110) — the flash-reducing moves — at a spot that *kept $S/\sqrt{B}$* at the 3.90 ceiling. Same sob, -5.8% flash.

Central-hole A/B — reproduces Edmonds

Matched high-stats A/B: solid target vs a **21.5 mm** central hole (~ 40 M input e^- each).

observable	solid	holed	hole effect
flash-event rate	—	—	−24%
total flash per POT	8.2×10^{-7}	6.4×10^{-7}	−21.5%

- Same direction and comparable magnitude to **Edmonds DocDB-10898** ($\sim 30\%$) — validates the flash pipeline end-to-end.
- The **deployed** Run1 target already carries this hole.

Honest reading of the improvement

- **Flash: a tie at the floor, not a breakthrough.** Every “better-flash” point is $\leq 3\%$ below baseline — within run-level noise. The deployed 21.5 mm hole already sits at the flash floor.
- **$S/\sqrt{B}$: a real, resolved win.** Champion $+0.2$ ($\sim 20\sigma$) at the same flash floor — extra stopping mass buys more CE signal at no flash cost.
- **Caveat:** that gain carries un-modelled costs (added material, backgrounds this scan does not capture). Deploy-worthiness is a physics/cost call beyond the scan.

Conclusion & next steps

- Stopping-target foils are a **usable electron-flash lever**: $\approx 3.1\times$ swing in tracker flash per POT with **no $S/\sqrt{B}$ trade-off**.
- The **champion** beats the deployed target on both axes, and the $S/\sqrt{B}$ ceiling is measured and priced — **3.84 at +26% flash**.
- The central-hole A/B **reproduces Edmonds**, confirming the flash pipeline is physically sound.
- **Next:** the central-hole line (bo-foilshole) — Edmonds' $\sim 30\%$ central-hole lever is an order of magnitude larger than the outer envelope bought.